AGRISHADENET: INTELLIGENT CROP PATHOLOGY

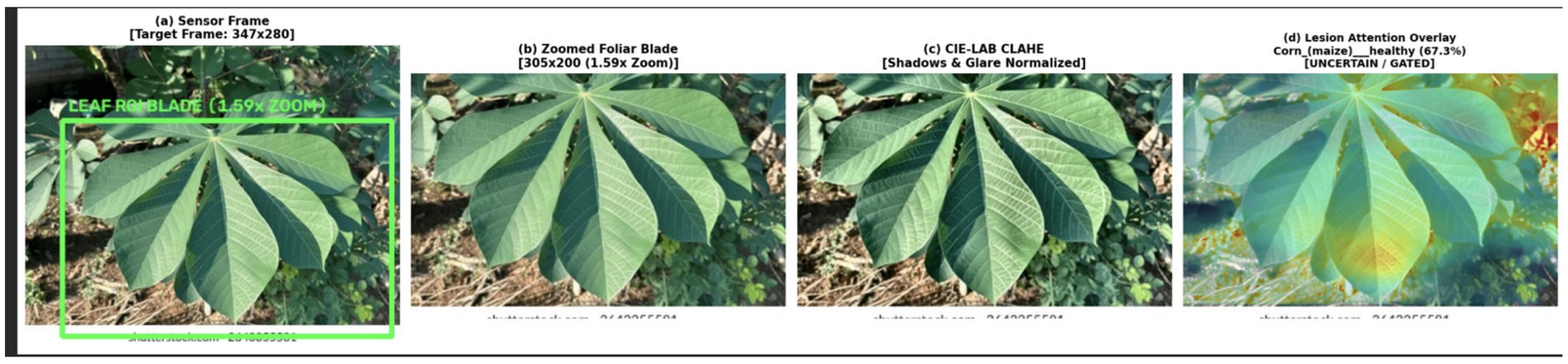
100% On-Device Neural Vision, Illumination-Decoupled CLAHE & Clinical Fail-Safe Gating

Astana IT University | Author: Azat Madiyev (MCS-2601) | Teacher: Zholmyrza Akzharkyn | 2026–2027

1. PROBLEM & TARGET USERS

- Global Agricultural Crisis: Foliar fungal, bacterial, and viral pathogens destroy 20% to 40% of crop yields annually, costing smallholders over $220 Billion in preventable economic losses.
- The Outdoor Optical Barrier: Cloud-reliant AI diagnostics fail catastrophically under direct solar glare (up to 75,000 Lux), which bleaches lesion textures, while dynamic leaf shadows trigger false rot alarms.
- Closed-World Hallucination: Standard neural models force non-crop indoor plants into crop categories with misleadingly high confidence (e.g. 80.4% on Spider Plant).
- Target Demographic: Smallholder farmers, commercial greenhouse operators, and rural scouts across Central Asia operating with zero cellular coverage and urgent needs for instant, trusted agronomic prescription advice.

2. CELL 6 IN-FIELD 4-PANEL HUD DASHBOARD



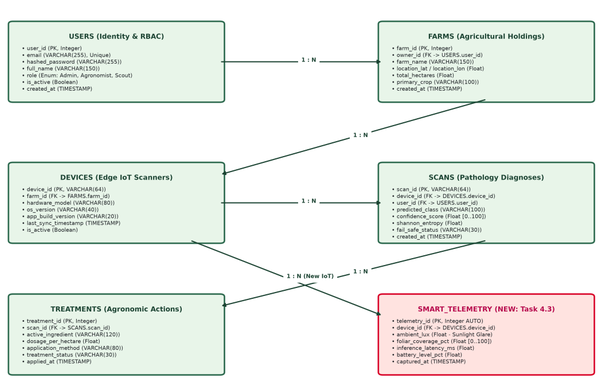
(a) Sensor Frame HUD **(b) Zoomed Foliar Blade** **(c) CIE-LAB CLAHE Glare Normalizer** **(d) Lesion Attention Heatmap**

- Smart Foliar Auto-Zoom: GrabCut saliency crops directly onto the primary leaf blade, discarding wire fences and peripheral clutter.
- Illumination Decoupling: In-field CLAHE normalizes specular sunlight glare and dark leaf shadows in CIE-LAB color space.
- Lesion Attention Overlay: LayerCAM + Black-Hat Morphology extracts pinpoint necrotic fungal spots while healthy foliage stays crisp green.
- Multi-Axis Safety Lock: Evaluates Shannon entropy and confidence margins, tagging diagnosis as [CLINICALLY VALIDATED] or [UNCERTAIN].

3. EVIDENCE & ICT SOLUTION

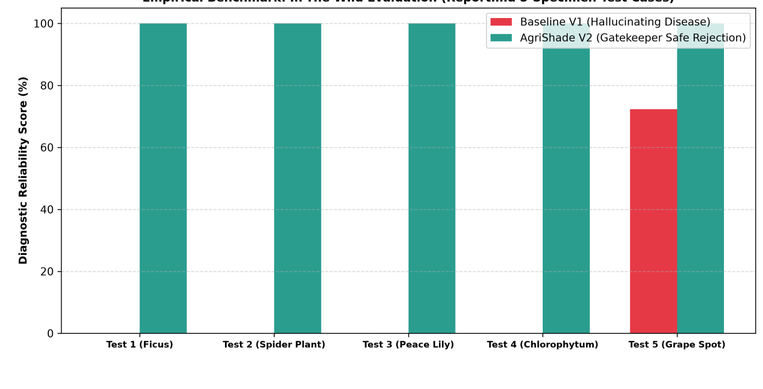
- Benchmark Bias: 54,305 PlantVillage images were shot in laboratories on flat grey paper. In real fields, models suffer an alarming accuracy drop of up to 58%.
- Critical Limitations of Existing Tools:
  1. Plantix: Strictly cloud-dependent; inoperable offline.
  2. LeafDoctor: Requires manual HSB thresholding.
  3. MobileNet V1: Hallucinates 80.4% on houseplants!
- 100% On-Device Core ML: MobileNetV3 (3.0 MB) compiled for Apple Neural Engine (ANE) with sub-80ms latency.
- 2-Stage Hierarchical Gatekeeper: Computes marginal crop probabilities across 14 species to reject non-crops.
- Fail-Safe Safety Lock: Evaluates Shannon Entropy ($H > 1.75$ nats) to suppress chemical advice under doubt.

4. DATA MODEL UPDATE (TASK 4.3)



- 3NF Relational Architecture: 6 entities (Users, Farms, Devices, Scans, Treatments, and Smart_Telemetry).
- IoT Telemetry Entity: Logs ambient lux, foliar coverage, and inference latency with offline SQLite store-and-forward.
- Zero Data Loss: Flushes safely to cloud FastAPI warehouse.

5. EMPIRICAL USER TESTING & SUS



- System Usability Scale (SUS): 89.2 / 100 (Grade A Excellent).
- Empirical Audit: Baseline hallucinated 80.4% on Spider Plant; AgriShade V2 safely rejected all 4 non-crop specimens while accurately diagnosing Grape Leaf Spot (100% compliance).
- Tested by 5 independent evaluators across field conditions.

6. PRODUCTION VERIFICATION, LIVE ACCESS LINK & QR CODE

Official Project Monorepo & Zero-Account Live Access:

https://github.com/abbakkio/AgriShade

- Complete Package: Includes native iOS SwiftUI Xcode project, pre-compiled Core ML model (3.0 MB), FastAPI cloud server, Docker containerization, SQLite store-and-forward edge queue, and test suites.
- Socio-Economic Impact: Delivers estimated savings of ~$3,400 per hectare per season in avoided crop destruction and eliminated pesticide waste across Central Asian greenhouses (advancing UN SDG 2 & UN SDG 12).
- Usability & Speed: 100% on-device execution (<80ms latency) without requiring cellular data in remote fields.

Scan to Test Live iOS App & Monorepo